

KONA YA HISABATI – WEB DEVELOPMENT GUIDELINE & FLOW-CHART

A. DEVELOPMENT GUIDELINE

1.0 Background Summary for Developers

Kona Ya Hisabati is a digital version of the classroom Mathematics Learning Corner. It aims to address the lack of well-equipped math corners in many schools by providing **online, child-friendly, interactive numeracy activities** accessible anytime, at school or home.

Target Users:

- **Pre-Primary pupils**
- **Standard One and two pupils**
- **Teachers**
- **Parents/Caregivers**

Platform Goals:

- Improve numeracy skills through interactive practice.
- Provide continuous access to curriculum-aligned math resources.
- Strengthen school–home collaboration.
- Boost children’s interest, engagement, and self-confidence in mathematics.

2.0 Functional Requirements

2.1 User Categories

1. Learner (Pupil)

- Access games and activities.
- Receive feedback (audio + visual).
- Navigate using icons (minimal text).

2. Teacher

- Select activities for class.
- Monitor learner progress.
- Download simple worksheets.

3. Parent

- Monitor child’s performance.
- Support home practice.

4. Admin

- Upload content.
- Manage accounts/permissions.
- Review analytics.

3.0 Platform Structure & Modules

A. Home Page

- Child-friendly interface
- Colorful icons
- Audio-assisted navigation eg Tap here for Counting

B. Main Learning Modules

1. **Number Concepts**
Counting, number recognition, sequencing, tracing, missing numbers.
2. **Shapes & Patterns & Spatial Awareness**
Identify, sort, complete patterns, left/right, above/below.
3. **Basic Operations (Addition/Subtraction)**
Drag & drop, number lines, visual operations, word problems.
4. **Sorting, Matching, Classifying**
Sort by color/size, categorize objects, match numbers.
5. **Measurement Basics**
Big/small, tall/short, heavy/light, ordering sizes.
6. **Time & Daily Routine**
Day/night, morning/evening, sequencing activities.
7. **Money Concepts**
Identify coins/notes, match values, simple buying/selling.
8. **Math Games (Play Zone)**
Puzzles, matching games, memory cards, number hunts.
9. **Math Songs & Rhymes**
Counting songs, animations, shape songs.
10. **Quizzes & Assessment**
Auto-graded quizzes, badges, progress reports.

4.0 Technical Requirements

4.1 Front-End

- Framework: React.js / Vue.js (recommended)
- Responsive design for tablets and phones
- High-contrast colors and large buttons for children

4.2 Back-End

- Admin dashboard for content uploading

4.3 Accessibility

- Audio instructions for non-readers
- Minimal text
- Visual cues (arrows, icons)
- Offline mode (optional)

4.4 Security

- Child-safe authentication (no email needed for pupils)
- Teacher/parent accounts secured by password
- No personal data collection from children

5.0 Content Development Guidelines

1. Curriculum Alignment

- Align with pre-primary and Grade 1 numeracy standards in Tanzania.

2. Activity Design

- Simple instructions with voice support.
- Use bright colors, familiar objects (animals, fruits, everyday items).
- Immediate positive feedback (sounds, stars, animations).

3. Game Mechanics

- Drag-and-drop
- Matching
- Tap-to-select
- Audio prompts

4. Assessment

- Short quizzes after each topic
- Automatic scoring
- Generate simple reports for parents/teachers

6.0 Development Phases

Phase 1: Planning & Analysis

- Review curriculum
- Define modules
- Draft UI/UX wireframes

Phase 2: Content & Activity Design

- Prepare all visuals (icons, shapes, objects)
- Script audio instructions
- Prepare animations

Phase 3: System Development

- Build front-end interface
- Set up back-end, API, database
- Integrate content

Phase 4: Testing

- Child usability testing
- Teacher feedback
- Bug fixing

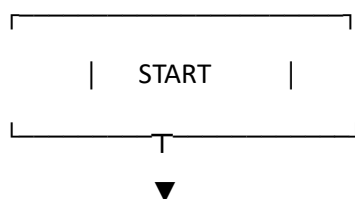
Phase 5: Deployment

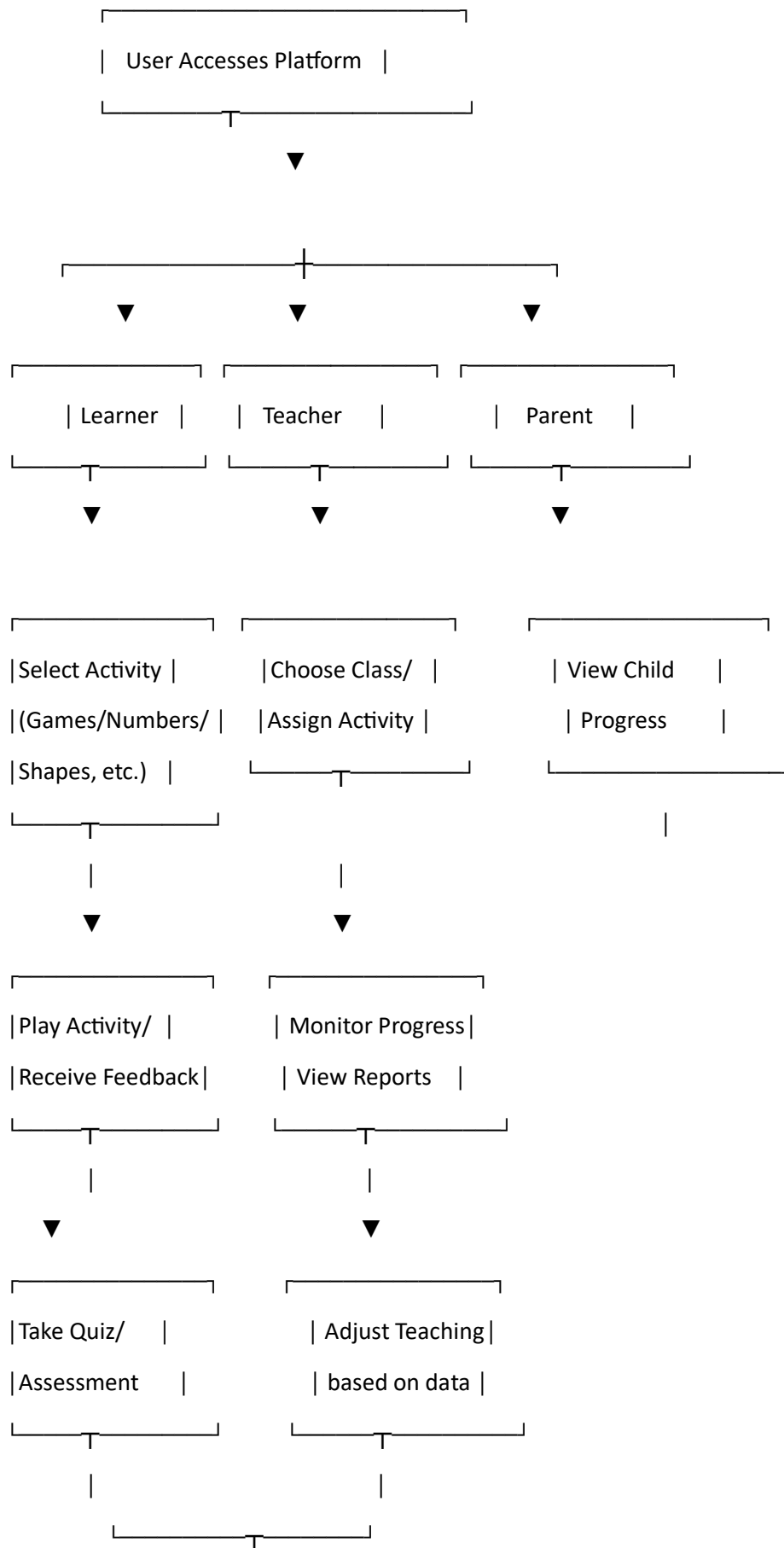
- Launch web platform
- Set up analytics

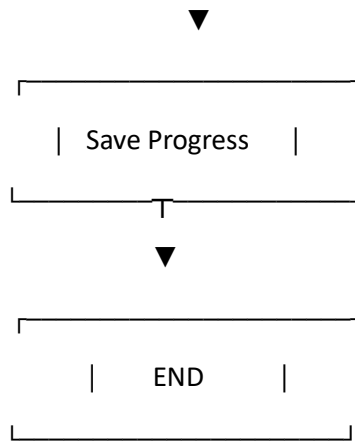
Phase 6: Continuous Improvement

- Add new games
- Update songs & quizzes
- Teacher feedback integration

7.0 Flow Chart Diagram







8.0 Deliverables for Developers

- Functional platform with full modules listed.
- UX optimized for children (3–7 years).
- Audio-assisted instructions.
- Teacher & parent dashboards.
- Secure, scalable architecture.
- Admin for adding new content.